ENGINEERING RESEARCH / UNPUBLISHED MANUSCRIPT DRAFT

Start and Stop Delay Constraints in EMG Command Filtering

A Reproducible Replay Benchmark

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THE ORIGINAL HYPOTHESIS WAS UNSUPPORTED.

No tested development family met the joint requirements.

01 The engineering question

Can separate start and stop persistence reduce unwanted commands during annotated rest while preserving timely, sustained movement commands?

Suppressing short bursts can delay both starts and stops.

02 What was built

01 Decode
Personalized LDA; 150 ms trailing windows

02 Filter
Seven causal controller families

03 Replay
Hold each command until the next decision

04 Score
Timing, rest exposure, and correct/wrong output

25 ms hops. Stable confirmation spans 100 ms and is recorded at the confirming endpoint, never backdated.

03 Requirements assessed jointly

START

At least 90% stable-correct onsets by 250 ms

STOP

At least 90% stable-rest confirmations by 150 ms

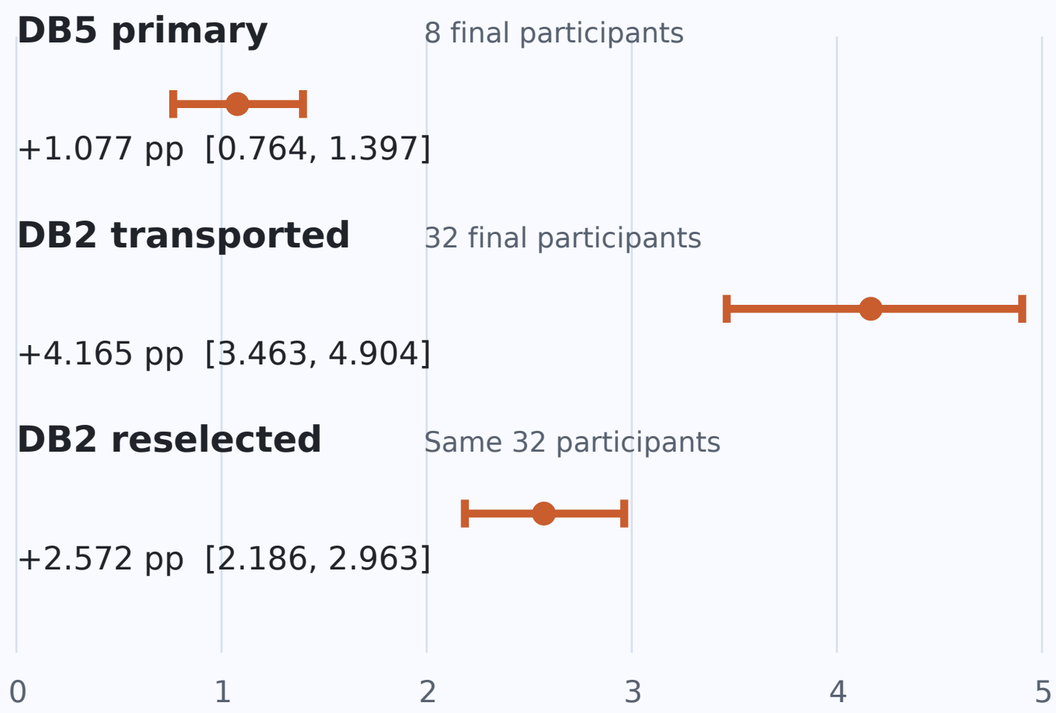
COVERAGE

At least 80% nonzero active-interior coverage

Coverage includes wrong commands. It is not accuracy. The active interior begins 500 ms after onset.

04 Actual paired comparisons

Asymmetric minus symmetric persistence
Change in command exposure during rest (percentage points)



Points: paired mean differences. Lines: descriptive paired bootstrap 95% intervals. Positive values mean more unwanted activation during annotated rest.

0/8 DB5 0/32 DB2

Participants meeting all requirements, for every evaluated method: 14 DB5 methods and 19 DB2 methods.

05 What the result means

The contribution is a reproducible benchmark and an explicit account of the tradeoffs, rather than a new control algorithm or a demonstrated improvement.

The study includes 23 prespecified sensitivity cases and a DB2 replication at its native sample rate. Paired contrasts remain descriptive after failed development feasibility.

SCOPE & LIMITS

Offline intact-limb recordings, not a clinical or closed-loop prosthetic trial. Retrospective annotations do not measure intention. Dataset IDs do not establish independent biological cohorts. No journal publication is claimed.

Paper, code, poster, and interactive lab

ksricharan7.github.io/portfolio/emg.html

Source: Ninapro DB5 and DB2; released benchmark outputs. Software release v0.11.0.